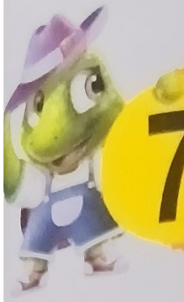

7 Decimals

Tune In

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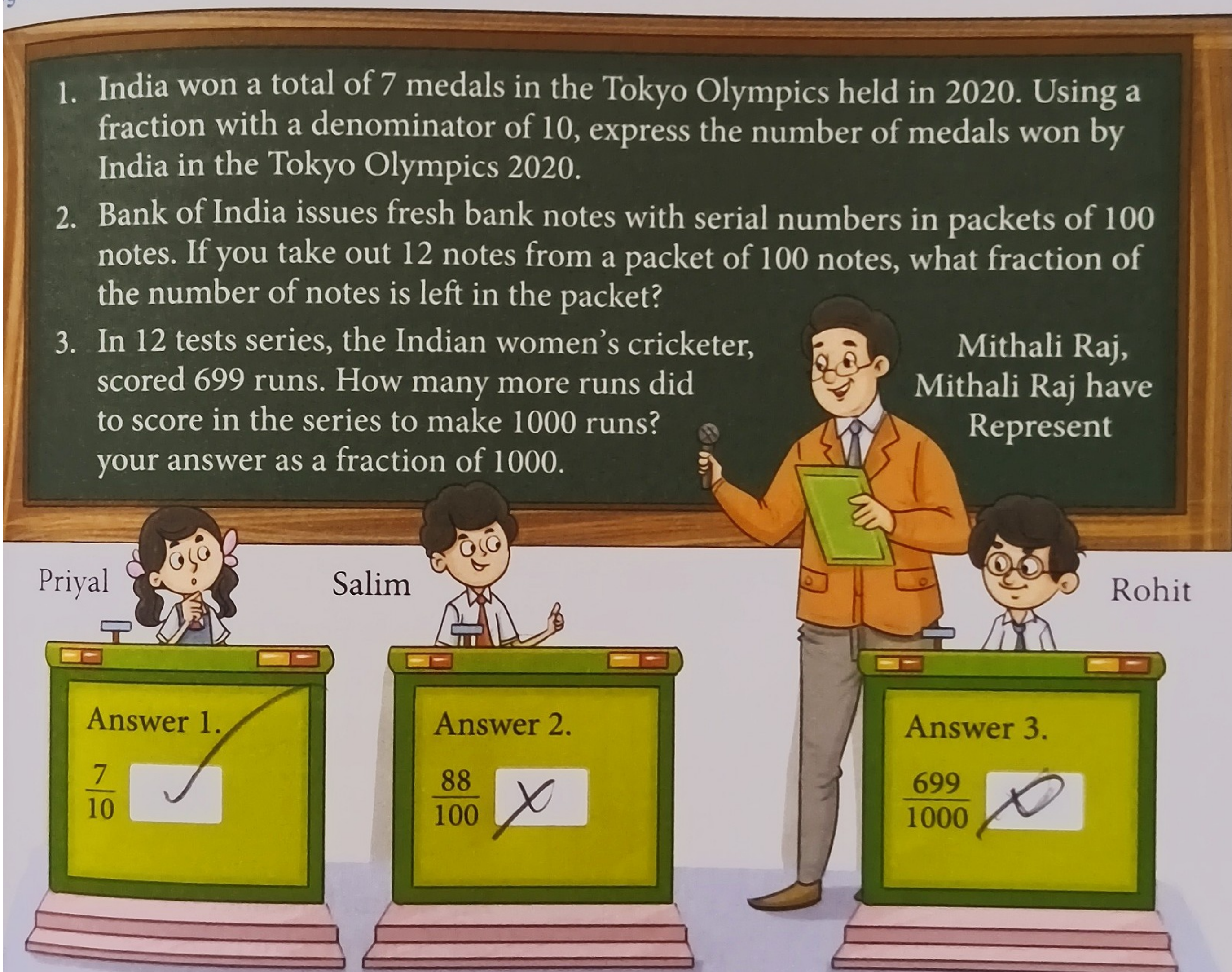
21st CENTURY SKILLS
KNOWLEDGE OF INDIA

quiz show is designed to test a person's memory, knowledge, agility, or luck. Priyal, alim, and Rohit, grade IV students, participated in a quiz show held in their school. They ll scored the same points. So, in the end, their teacher asked them the last question to ach of them shown on the board. Read the questions and check the answers below given y them. Tick the correct answer and cross the incorrect one in the given boxes.



INTRODUCTION TO DECIMALS

Look at the correct answers you found above.

$\frac{7}{10}$ $\quad$ $\frac{88}{100}$ $\quad$ $\frac{301}{1000}$

What do you observe?

We observe that the fractions have 10 or the product of two or more 10s as the denominators.

100 is the product of two 10s, that is, $100 = 10 \times 10$.

1000 is the product of three 10s, that is, $10 \times 10 \times 10 = 1000$.

Such fractions are known as **decimal fractions**.

We shall now learn more about fractions of this type.

Decimal fractions are fractions whose denominator is 10 or a product of two or more 10s. These can be written in a different form using a decimal point (.).

Let's Try! 1

Write each fraction as a decimal number.

1. $\frac{1}{10}$ $\quad$ 2. $\frac{7}{100}$ $\quad$ 3. $\frac{95}{1000}$

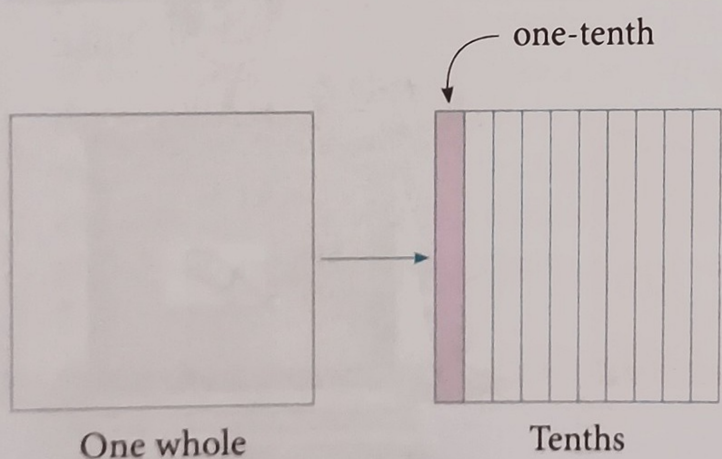
$\frac{7}{10}$ is written as 0.7. It is read as 'zero point seven'.

$\frac{88}{100}$ is written as 0.88. It is read as 'zero point eight eight'.

$\frac{301}{1000}$ is written as 0.301. It is read as 'zero point three zero one'.

Tenths

When one whole is divided into 10 equal parts each part is called one-tenth.



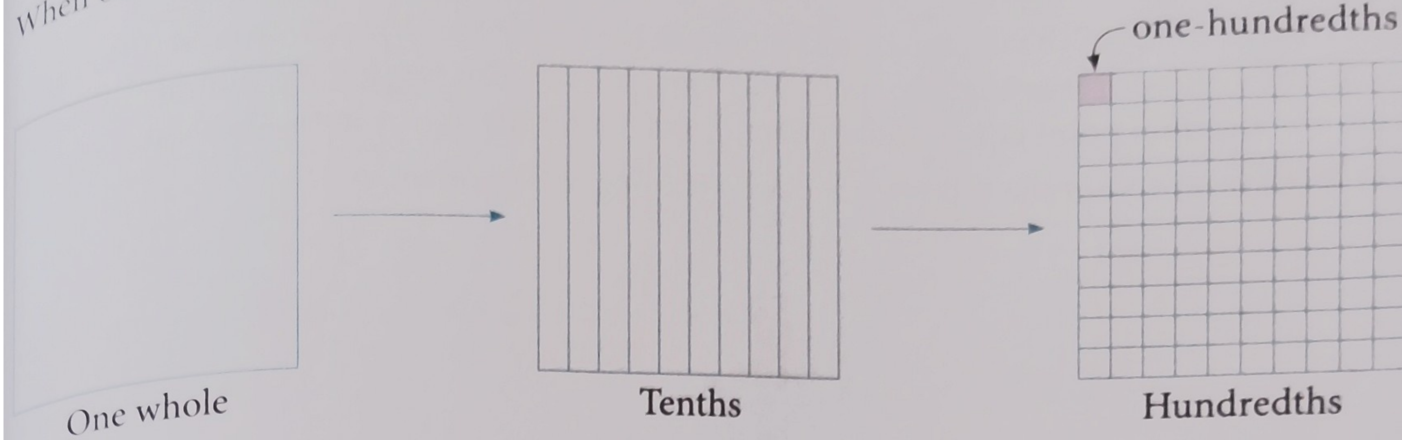
The shaded part can be written as a fraction or a decimal.

Fraction	Decimal
$\frac{1}{10}$	0.1

The decimal number 0.1 is read as one-tenth or zero point one.

Hundredths

When one-tenth is further divided into 10 equal parts, each part is called one-hundredth.



The shaded part in the above square can be written as a fraction or a decimal as shown below.

Fraction	Decimal
$\frac{1}{100}$	0.01

The decimal number 0.01 is read as one-hundredth or zero point zero one.

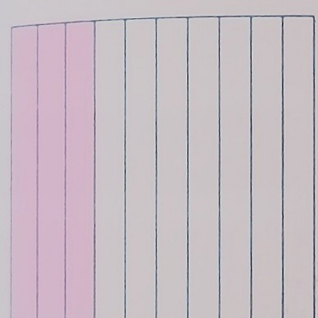
Watch Out!

One-hundredth

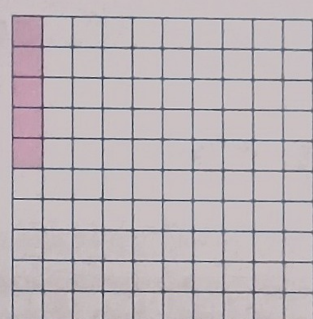
0.1 (✗) 0.01 (✓)

Example 1: Express the following using fractions and decimals.

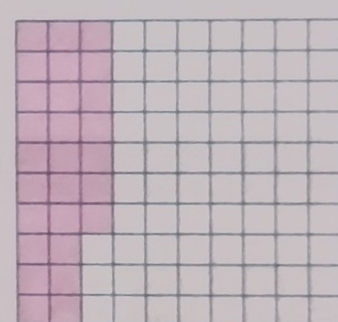
a.



b.



c.



Solution: a. In the given square 3 out of 10 parts are shaded.

It can be expressed as $\frac{3}{10} = 0.3$

b. In the given square 5 out of 100 parts are shaded.

It can be expressed as $\frac{5}{100} = 0.05$

c. In the given square 27 out of 100 parts are shaded.

It can be expressed as $\frac{27}{100} = 0.27$

In these decimals the whole number part is 0, so these decimal numbers are less than one.

Let's Try! 2

Shade to show the decimal.

1. 0.7

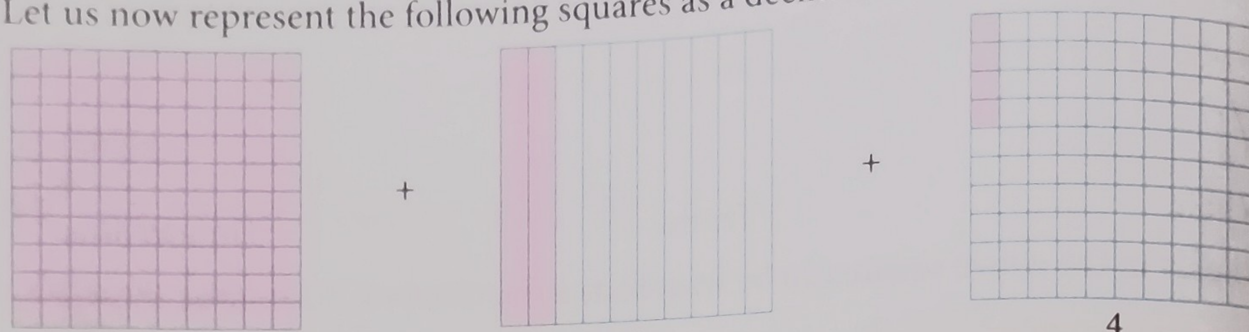
2. 0.4

Now, let us learn about decimal numbers greater than one.

In the mixed number $2\frac{44}{100}$, 2 is the whole number part and $\frac{44}{100}$ is the fractional part.

It is written as 2.44. The decimal point separates the whole number part from the fractional part. The number is read as 'two point four four'. We can also see that the number of digits after the decimal point equals the number of zeros after the digit 1 in the denominator.

Let us now represent the following squares as a decimal number.

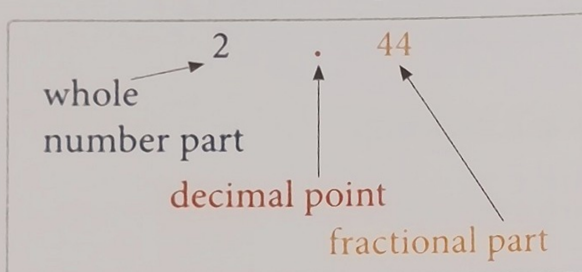


$1 \quad + \quad \frac{2}{10} \quad + \quad \frac{4}{100} \quad = \quad 1.24$

1.24 is read as one point two four or one and twenty-four hundredths.

Parts of a Decimal Number

A decimal number has two parts separated by a decimal point.



Watch Out!

0.08 is eight-hundredth (✗)

0.08 is eight-hundredths (✓)

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CREATING

Exercise 7A

Go to MyInspire and attempt questions 1 and 2.

3. Show the following fractions by shading the grids. Also, write each decimal fraction as a decimal number.

   a. $\frac{2}{10}$ = 0.2

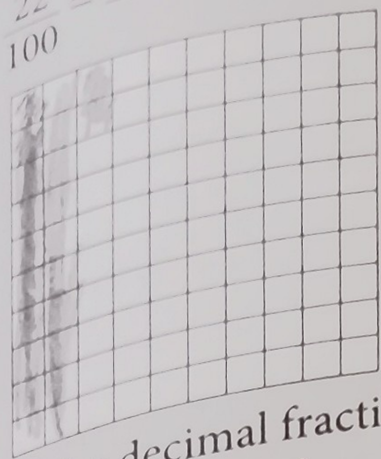
   b. $\frac{5}{10}$ = 0.5

   c. $\frac{7}{10}$ = 0.7

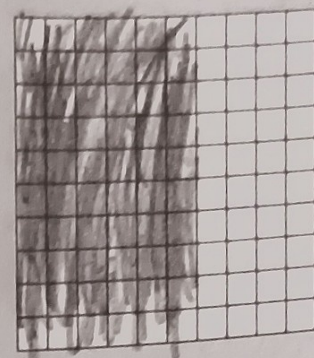
Teacher's Note

The students identify that the zeros in the denominator of a decimal fraction tells us the number of digits after the decimal point in the decimal number.

d. $\frac{22}{100}$ = 2.22



e. $\frac{60}{100}$ = 0.60



Write the decimal fractions as decimal numbers.

a. $\frac{4}{10}$ b. $\frac{3}{100}$ c. $\frac{16}{100}$ d. $\frac{145}{1000}$ e. $\frac{2499}{1000}$

f. $\frac{38}{10}$ g. $\frac{463}{100}$ h. $\frac{47}{1000}$ i. $\frac{6}{1000}$ j. $\frac{256}{100}$

Write the decimal numbers as decimal fractions.

a. 5.8 b. 0.9 c. 0.17 d. 0.123 e. 0.037

f. 0.004 g. 1.12 h. 0.058 i. 0.03 j. 4.82

Write the decimals in words.

a. 0.02 b. 1.5 c. 16.15 d. 7.4 e. 8.34

Complete the table.

Fraction	$\frac{7}{10}$	$\frac{7}{100}$			$6\frac{7}{100}$	
Decimal	0.7		1.7	5.77		2.6

PLACE VALUE

The place value of the first digit to the right-hand side of the decimal point is got by dividing it by 10 (tenth).

The place value of the second digit to the right-hand side of the decimal point is got by dividing it by 100 (hundredth).

This pattern can be continued to the third place (thousandth) and so on. The place value chart is now modified to include the fractional part.

... the number 5134.1769.

		...al point	...nths	Hundredths	Thousandths	Ten thousandths
					6	9

In the above example, the first digit after the decimal point has the place value one-tenth $(\frac{1}{10})$, the second digit after the decimal point has the place value seven-hundredths $(\frac{7}{100})$, the third digit has the place value six-thousandths $(\frac{6}{1000})$.

Starting from the first digit to the right-hand side of the decimal point, each digit gets divided by 10, 100, 1000, and so on.

Let's Try! 3

Write the place value of 7 in the following.

1. 2.175 2. 54.706 3. 0.397

Expanded Form

Consider the number 381.49.

The expanded form is $300 + 80 + 1 + \frac{4}{10} + \frac{9}{100}$.

Let us write the expanded form of the following numbers.

Number	Expanded form
0.024	$\frac{2}{100} + \frac{4}{1000}$
0.157	$\frac{1}{10} + \frac{5}{100} + \frac{7}{1000}$
7.2	$7 + \frac{2}{10}$
0.006	$\frac{6}{1000}$

Let's Try! 4

Write the following numbers in the expanded form.

1. 302.839 2. 2932.302

When we are given the expanded form of a number, we can write the number in the **standard form** by observing the place values.

$$3000 + 40 + 6 + \frac{1}{10} + \frac{9}{100} + \frac{5}{1000} = 3046.195$$

LIKE AND UNLIKE DECIMALS

Decimals having the **same number of digits after the decimal point** are called **like decimals**. If they do not have the same number of digits after the decimal point, then the decimals are called **unlike decimals**.

Like decimals
3.162, 0.997, 5.324, 8.105

Unlike decimals
0.8, 1.88, 0.344, 9.0166

Converting Unlike Decimals to Like Decimals

Now, consider 7.4, 7.40, and 7.400.

Writing them as fractions, we get $\frac{74}{10}$, $\frac{740}{100}$, and $\frac{7400}{1000}$.

We can see that these three fractions are equivalent, that is, $\frac{74}{10} = \frac{740}{100} = \frac{7400}{1000}$.

So, 7.4 = 7.40 = 7.400.

...at is, adding zeros at the right end of the fractional part of a ...cimal number does not change the value of the number.

...is idea is used to change unlike decimals into like decimals.

...e unlike decimals given above are 0.8, 1.88, 0.344, and 9.0166. ...0166 has the greatest number of decimal digits which is 4. ...writing the other three numbers with 4 decimal places, ...get the equivalent decimals 0.8000, 1.8800, and 0.3440.

...8000, 1.8800, 0.3440, and 9.0166 are like decimals.

Unlike decimals can thus be changed into like decimals by adding the required number of zeros to the right end of the fractional part of the number.

Go to MyInspire for

Key Point

Adding zeros at the right end of a decimal number does not change the value of the number.

Let's Try! 5

1. Circle the like decimals.
   0.5, 1.07, 2.1, 5.7, 6.4, 0.008
2. Convert into like decimals.
   8.8, 0.88, 5.087

...ercise 7B

Go to MyInspire and attempt question 1.

Write the expanded form.

a. 9.59 b. 45.2 c. 456.809 d. 0.329

Write the numbers in standard form.

a. $7 \times 10 + 2 \times 1 + \frac{1}{10} + \frac{6}{100} + \frac{1}{1000}$

b. $4 \times 100 + 2 \times 10 + \frac{1}{10} + \frac{7}{1000}$

c. $9 \times 1000 + 8 \times 10 + \frac{3}{100} + \frac{2}{1000}$

d. $5 \times 10 + \frac{1}{10}$

Form three groups of like decimals.

1.1, 2.54, 1.08, 0.897, 2.305, 4.6, 3.45, 23.4, 0.99, 1.95, 0.7, 6.566

Change all the numbers into like decimals.

2.7, 78.56, 34.088, 1.234, 5.666, 4.5, 2.09, 8.98

Let's Review

3/12/25

Revision Exercises

Go to MyInspire for

BLOOM'S REVISED TAXONOMY

Start Up

REMEMBERING UNDERSTANDING

1. Complete the table.

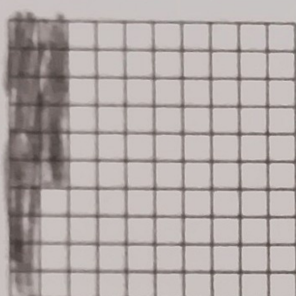
Fraction	$\frac{34}{1000}$			$\frac{523}{100}$	$\frac{7108}{1000}$	
Decimal		19.609	3.95			3.766

2. Write the fractions and decimals for the shaded part.

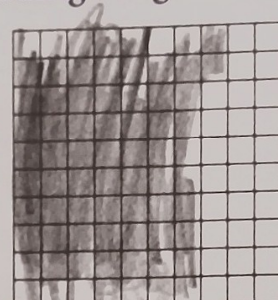
0.3 a. $\frac{3}{10}$ b. 0,6 $\frac{6}{10}$

0,5 c. $\frac{5}{10}$ d. 0.7 $\frac{7}{10}$

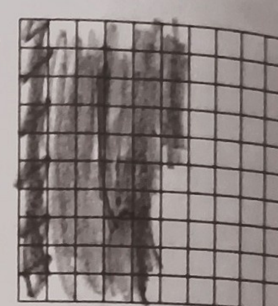
3. Show the decimal numbers by shading the grids.

a. 

0.16

b. 

0.72

c. 

0.55

4. Convert the given unlike decimals to like decimals.

a. 3.12, 8.1, 6.908 b. 55.9, 6.188, 4.88 c. 224.0, 7.122, 19.3 d. 40.73, 8.8, 99.55

Level Up

APPLYING ANALYSING

1. Write as decimals.

a. 7 tenths b. 8 hundredths c. 3 thousandths

d. 2 tenths 5 hundredths e. 16 thousandths f. 9 and 25 hundredths

2. Write the place value of the coloured digits.

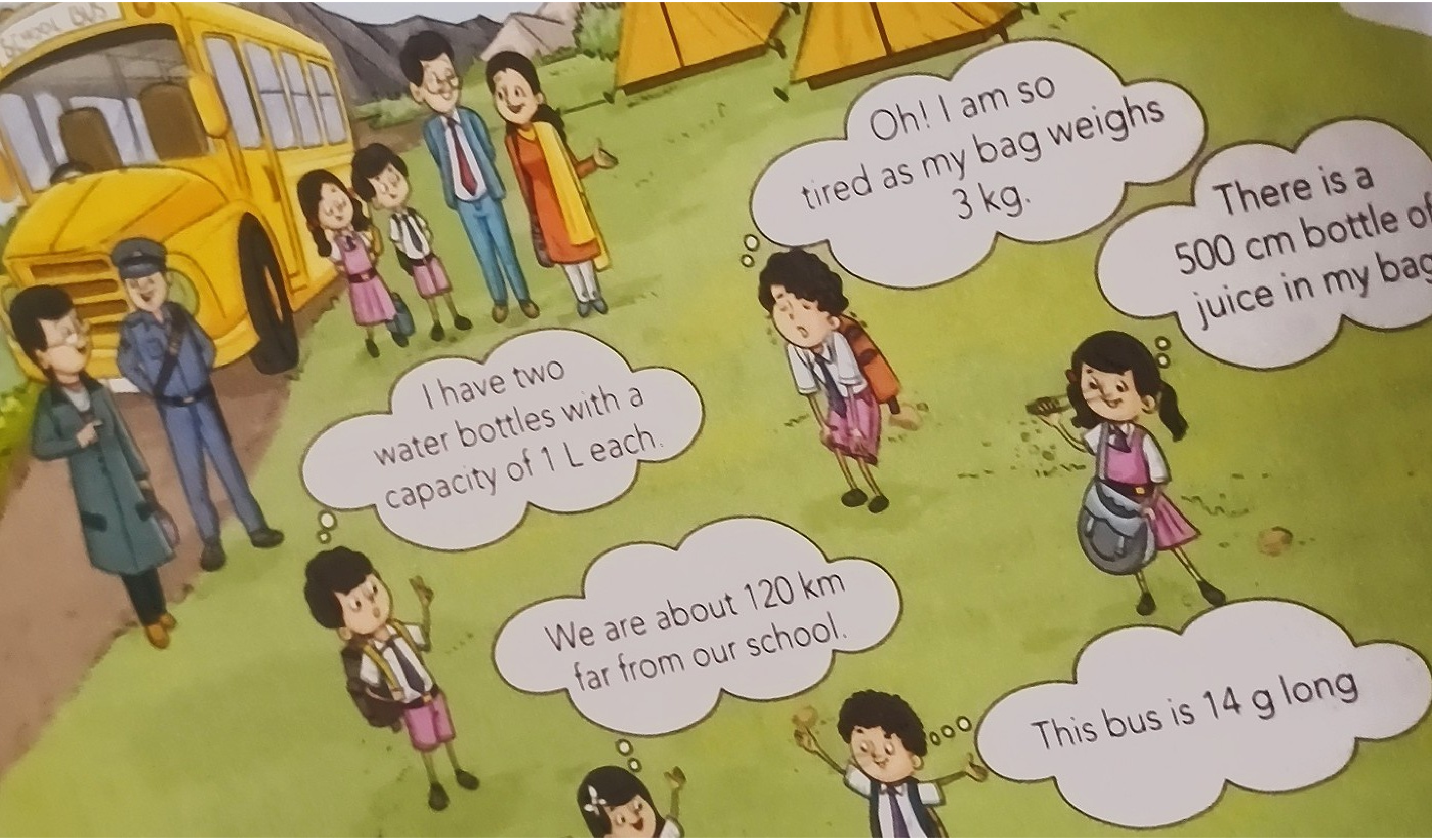
a. 15.67 b. 0.129 c. 23.245 d. 1.087

e. 0.85 f. 55.2 g. 6.108 h. 9.1234

3. Which of the following are pairs of like fractions?

a. 0.5 and 0.05 b. 1.75 and 1.57 c. 25.6 and 2.56

d. 22.81 and 2.28 e. 9.012 and 0.912 f. 1.234 and 12.34



Oh! I am so tired as my bag weighs 3 kg.
There is a 500 cm bottle of juice in my bag
I have two water bottles with a capacity of 1 L each.
We are about 120 km far from our school.
This bus is 14 g long